PROJECT BLUEPRINT · PRODUCT & ENGINEERING PLAN

Apollo Elevators

Digital Platform & Elevator Service Management System (ESMS)

From Marketing Website to Full-Stack ERP/CRM — Architecture, Workflows, Sitemap & Sprint-Ready User Story Backlog

PREPARED FOR	Apollo Elevators — Bengaluru, Karnataka
STACK	Next.js 15 · Spring Boot 4 (Java 21) · PostgreSQL · MongoDB · AWS
SCOPE	Phase 0 – Phase 5 · Website → Admin → Service Mgmt → Engineer App → Customer Portal
DOCUMENT TYPE	Product Requirements, IA, Architecture, Workflows & Agile Backlog

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1. Executive Summary & Project Goals

Apollo Elevators currently has a brochure-style print catalog. This document restructures that content into a phased, sprint-ready digital product: it begins as an SEO-driven lead-generation website and evolves into a full Elevator Service Management System (ESMS) covering sales, AMC, breakdown service, field engineers and customer self-service — without needing an architectural rewrite at any stage.

Business Goals

- Rank on Google for high-intent local searches (e.g. "lift installation Whitefield", "elevator AMC Electronic City").
- Convert website visitors into qualified enquiries with minimal friction (call / WhatsApp / form).
- Replace manual, paper-based AMC and complaint tracking with a searchable digital system.
- Give field engineers a mobile-first tool to close jobs with photo/signature proof, eliminating disputes.
- Give customers self-service visibility into AMC status, service history and complaint tracking, reducing inbound calls.

Guiding Principles for Build-Out

- **Modular monolith first.** One Spring Boot codebase organized into clean modules (packages). Split into microservices only if scale ever demands it.
- **Relational core.** Customers, buildings, lifts, AMC, complaints and invoices are highly relational — PostgreSQL is the system of record. MongoDB is used only for logs, notifications and analytics/search cache.
- **Ship in phases.** Every phase (0–5) is independently deployable and delivers real business value on its own — the company can start using Phase 1–3 while Phase 4–5 are still in development.
- **User-story sized work.** Every feature in this document is broken down to a size a single developer can pick up, build and test within 1–3 days.

How to Use This Document

Sections 1–7 give the shared context (why, for whom, and how the system is architected). Sections 8–12 are the actual backlog — organized as Epics, each broken into User Stories with a priority (MoSCoW: Must / Should / Could) and acceptance criteria, ready to be copied as-is into Jira, Linear, Trello or GitHub Projects. Section 13 defines when a story is considered "done" and the non-functional requirements that apply across all phases.

2. User Personas & Roles

Five distinct roles use the platform across its lifetime. Each phase of the backlog is written from the point of view of one or more of these personas.

Website Visitor / Prospective Customer
Summary: A facility owner, builder, architect or homeowner researching elevator options, often comparing 3–4 vendors.
Goals: Quickly find the right lift type for their building, see proof of past work, and get a quote without calling.
Pain points today: Old site looks like a static catalog; no easy way to compare products or request a quote per-location.

Sales / Admin Staff
Summary: Apollo Elevators office staff who handle enquiries, prepare quotations and manage day-to-day CMS content.
Goals: See all enquiries in one place, respond fast, track quote status, and update the website without a developer.
Pain points today: Enquiries currently arrive by phone/WhatsApp with no central log; website content changes need a developer.

Operations Manager
Summary: Oversees AMC contracts, complaint SLAs, engineer allocation and revenue reporting.
Goals: See AMC renewals due, overdue complaints, and engineer performance at a glance.
Pain points today: AMC expiry and service history tracked in spreadsheets — renewals are missed, no SLA visibility.

Field Engineer / Technician
Summary: Visits sites to install, maintain or repair elevators; usually on a phone, sometimes with weak connectivity.
Goals: See today's job list, get directions, complete a checklist quickly, and avoid carrying paper job sheets.
Pain points today: Paper checklists get lost; no photo/signature proof of work; disputes over what was actually done.

Building Customer / Facility Owner (post-sale)
Summary: An existing customer with an installed lift, wants visibility without calling the office.
Goals: Check AMC validity, raise a complaint with a photo, download invoices/certificates, track a technician's visit.
Pain points today: No visibility into AMC status or service history; must call to check anything, including invoice copies.

3. Information Architecture — Website Sitemap

Phase 1 ships as a ~45-page SEO-focused site rather than a 5–6 page brochure. Products, services and localities each get a dedicated, indexable page so Google can match specific search intent (e.g. a visitor searching "lift AMC HSR Layout" lands on a page written for exactly that query).

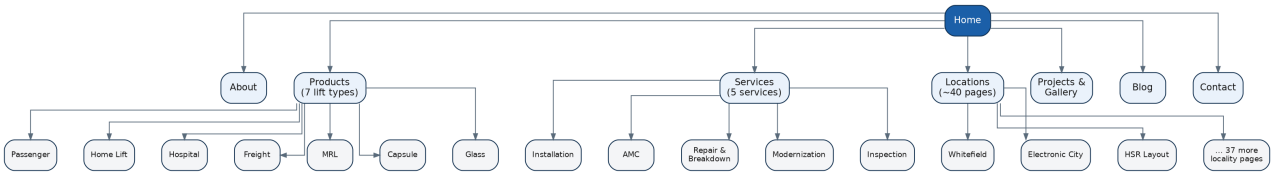


Fig 3.1 — Phase 1 site structure (products, services and ~40 locality pages collapse the same pattern)

Page Group	Page Count	Purpose
Home	1	Hero, trust signals, quick links to products/services, primary CTA (Get Quote)
About	1	Company history, certifications, standards compliance (IS 14665)
Products	7 + 1 index	One page per lift type: Passenger, Home, Hospital, Freight, MRL, Capsule, Glass
Services	5 + 1 index	Installation, AMC, Repair & Breakdown, Modernization, Inspection
Locations (SEO)	~40	"[Service] in [Locality]" landing pages for Bengaluru + city pages for Karnataka
Projects & Gallery	1–2	Case studies by building type: apartment, villa, hospital, commercial, industrial
Blog	1 index + posts	Long-tail SEO content, safety tips, maintenance guides
Contact / Legal	4	Contact, Privacy Policy, Terms, 404

4. System Architecture

A single Spring Boot backend (modular monolith) serves four Next.js frontends through one REST API. PostgreSQL is the system of record; MongoDB is used only for high-volume, low-structure data (logs, notification queues); S3-compatible storage holds all binary assets.

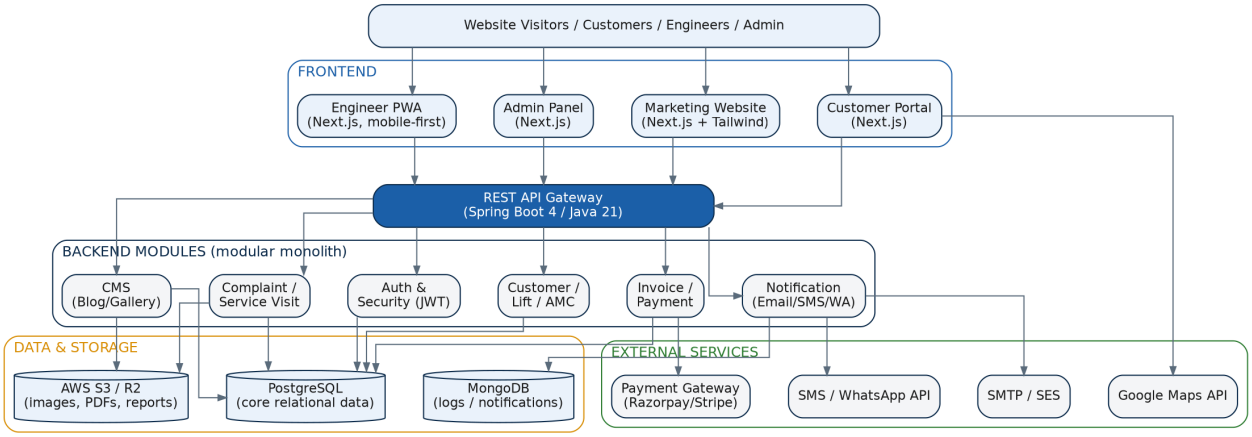


Fig 4.1 — Target architecture from Phase 2 onward (Phase 1 uses only the Marketing Website + CMS module + PostgreSQL)

Backend Modules (packages inside one deployable)

Module	Responsibility	Introduced In
auth	Login, JWT issuance/refresh, role-based access control	Phase 2
cms	Enquiries, blog, gallery, testimonials, SEO metadata	Phase 1 / 2
customer / lift / amc	Customers, buildings, lifts, AMC contracts	Phase 3
complaint / service-visit	Breakdown tickets, scheduled visits, checklists	Phase 3 / 4
invoice / payment	Billing for AMC, repair, installation; payment gateway integration	Phase 3 / 5
notification	Email / SMS / WhatsApp dispatch, templates, logs	Phase 2+
storage	Signed upload/download URLs to S3/R2 for images, PDFs, reports	Phase 1+

Suggested Repository Layout

```
apollo-elevators/
■■■ frontend/            Next.js 15 app (website + admin + customer portal as route groups)
■   ■■■ app/(site)/       public marketing pages
■   ■■■ app/(admin)/      admin panel routes
■   ■■■ app/(portal)/     customer portal routes
■   ■■■ components/  hooks/  lib/  services/  types/
■
■■■ backend/             Spring Boot 4, Java 21 — modular monolith
■   ■■■ auth/  users/  customers/  lifts/  amc/
■   ■■■ complaints/  projects/  notifications/  reports/
■   ■■■ common/  config/
■
■■■ engineer-app/        Next.js PWA (can also live as a route group in frontend/)
■■■ storage/             S3 bucket structure (see Section 5)
■■■ database/            PostgreSQL migration scripts (Flyway/Liquibase)
```

5. Database Design — Entity Relationship Overview

PostgreSQL holds all core business entities in a normalized relational model. The chain below is the backbone of Phase 3 onward: an Enquiry becomes a Customer, a Customer owns Buildings, a Building contains Lifts, and every Lift accumulates AMC contracts, Complaints, Service Visits and Invoices over its life.

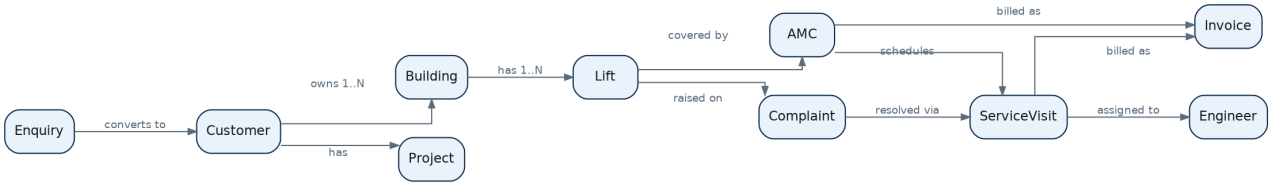


Fig 5.1 — High-level entity relationships (core relational chain)

Entity	Key Attributes	Notes
Customer	name, phone, email, type (individual/builder/facility)	Created from a converted Enquiry or added directly
Building	name, address, type, floors, locality	A customer can own multiple buildings
Lift	lift_no, capacity, stops, manufacturer, controller, install_date, warranty_end	The core asset every AMC/complaint/visit refers to
AMC	start_date, end_date, visits_per_year, status, contract_value	Drives the auto-scheduling engine (Section 6.2)
ServiceVisit	scheduled_date, type (AMC/repair), status, checklist_json, gps_start, gps_end	One row per engineer visit, linked to AMC or Complaint
Complaint	raised_date, description, priority, status, photo_urls	Raised by customer or admin; resolved via a ServiceVisit
Engineer	name, phone, skill_zone, active	Assigned to ServiceVisits
Invoice	amount, tax, due_date, status, pdf_url	Generated from AMC or ServiceVisit
Enquiry	name, phone, city, building_type, source, status	Captured from the public contact form
Project	type, status, start_date, handover_date	Represents a new installation from quote to handover

Where MongoDB Fits

- **Notification log** — every email/SMS/WhatsApp send attempt, payload and delivery status (high write volume, disposable).
- **Audit / activity log** — who changed what and when, across all modules (append-only, queried rarely).
- **Website analytics cache** — page views, search queries, funnel events feeding the admin dashboard.
- Everything else (customers, lifts, AMC, invoices, complaints) stays in PostgreSQL for referential integrity and reporting joins.

6. Core Business Workflows

These four workflows describe the operational backbone the backlog is built around. Each maps directly to a set of user stories in Sections 8–12.

6.1 Enquiry → Project Conversion (Phase 1 & 2)

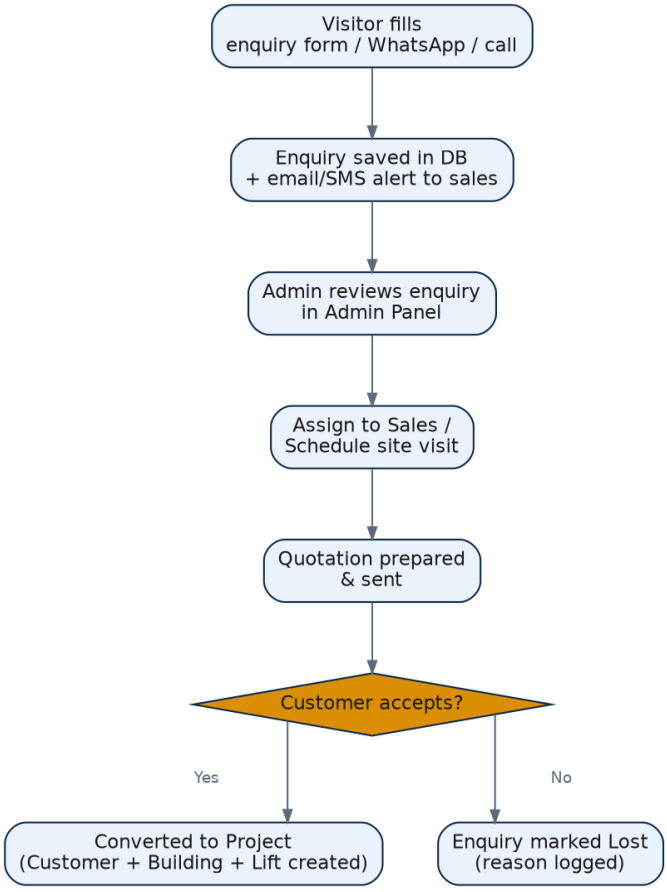


Fig 6.1 — Every website enquiry is tracked to a won/lost outcome; won enquiries seed the Customer/Building/Lift records used from Phase 3 onward.

6.2 AMC Lifecycle (Phase 3)

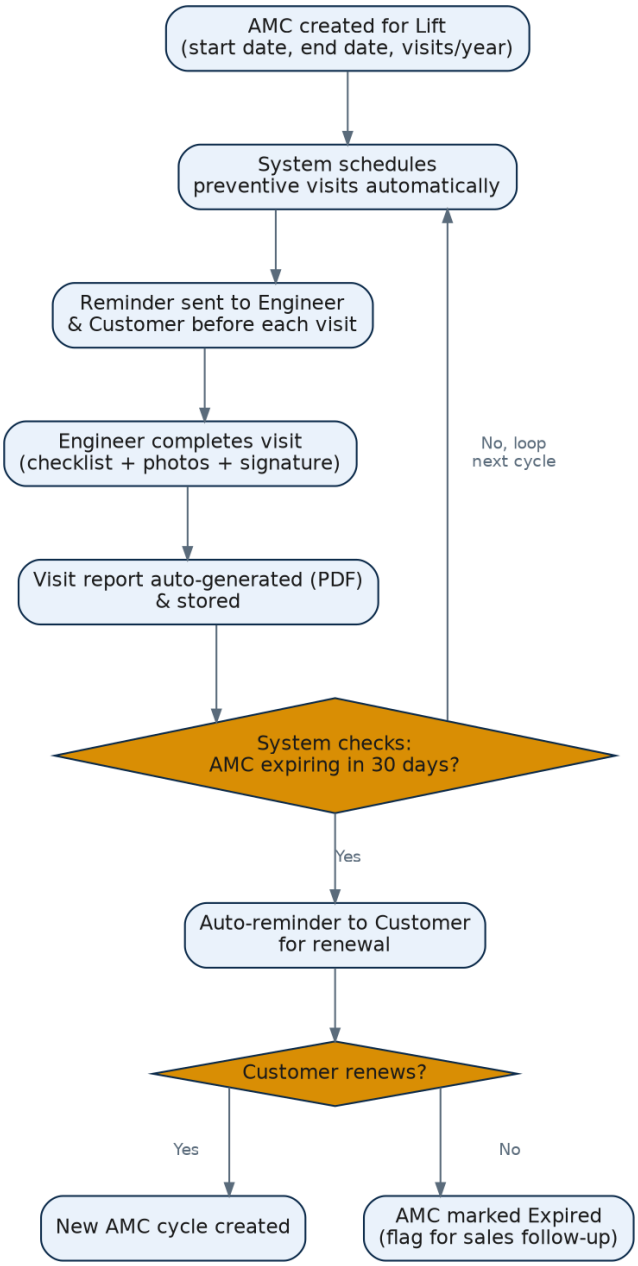


Fig 6.2 — AMC visits are auto-scheduled from the contract; renewal reminders fire automatically 30 days before expiry.

6.3 Complaint / Breakdown Resolution (Phase 3 & 4)

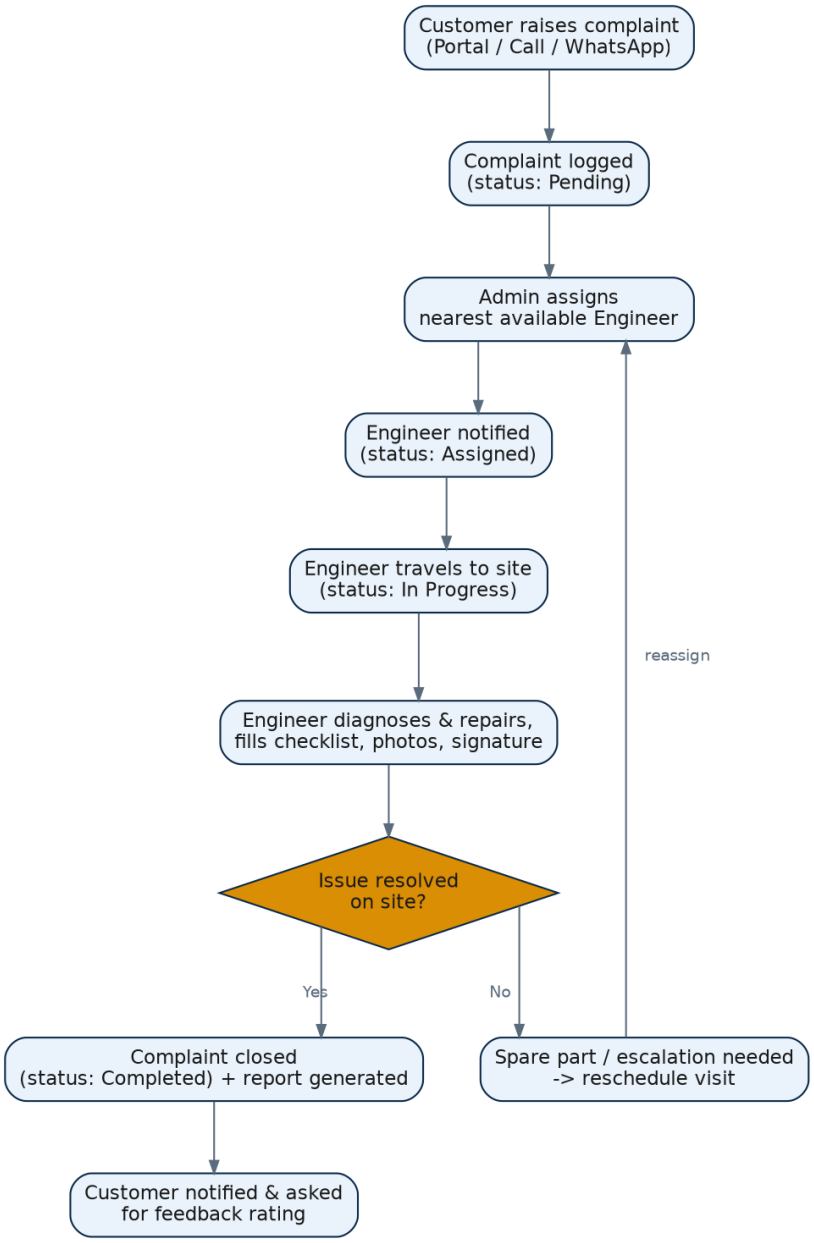


Fig 6.3 — Complaints loop back to reassignment if a visit can't be closed on the first attempt (e.g. spare part required).

6.4 Engineer's Daily Job Flow (Phase 4)

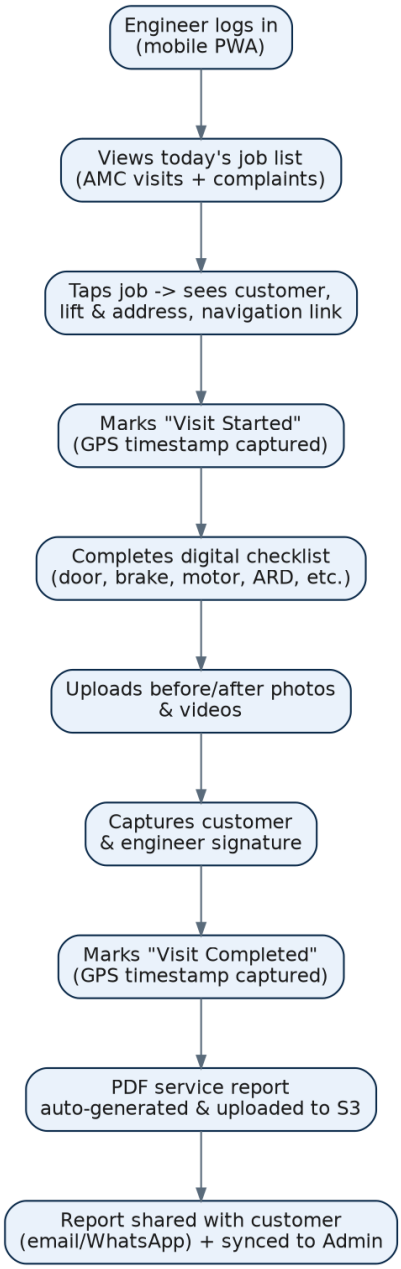


Fig 6.4 — GPS timestamps at visit start/end plus photo + signature capture remove all paper from field service.

7. Release Plan & Timeline

Each phase is a shippable release. Timelines assume one developer working part-time/full-time; compress if more engineering time is available.



Fig 7.1 — Sequential release timeline (Phase 0 – Phase 5)

Phase	Deliverable	Duration	Depends On
Phase 0	Discovery: content, brand assets, domain, hosting, repo setup	1 week	—
Phase 1	Public marketing website (~45 pages) + enquiry capture	2 weeks	Phase 0
Phase 2	Admin panel: auth, dashboard, enquiry mgmt, CMS, SEO tools	3 weeks	Phase 1
Phase 3	Service Management: customers, lifts, AMC, complaints, invoices	4 weeks	Phase 2
Phase 4	Engineer PWA: job list, checklist, photos, signature, GPS	3 weeks	Phase 3
Phase 5	Customer portal: self-service, online AMC renewal, feedback	4 weeks	Phase 3

Suggested Git Branching

`main → develop → feature/<module> (e.g. feature/auth, feature/amc, feature/engineer-checklist) → release/<phase> → hotfix/<id>`

8. Product Backlog — Phase 0 & Phase 1 (Marketing Website)

Goal: Generate qualified leads via an SEO-ready public website	**Duration:** 1 + 2 weeks	**Primary roles:** Sales/Admin (content), Visitor (consumer)

Phase 0 — Discovery (no code)

Epic 0.1 — Content & Asset Collection

#	User Story	Pri.	Acceptance Criteria
P0-1	As the project owner, I want a checklist of every asset needed (logo, brand colors, product photos, certificates, brochures, testimonials, past quotations/invoice formats) so nothing blocks Phase 1 development.	Must	Checklist created and shared; all assets received and stored in a shared drive before sprint 1 starts.
P0-2	As the project owner, I want the domain and hosting decided (apolloelevators.in/.com, AWS, GitHub org) so environments can be provisioned.	Must	Domain purchased; GitHub repo created with branch strategy; dev/staging/prod hosting accounts created.
P0-3	As the project owner, I want the full site copy (from the brochure rewrite) organized page-by-page so content entry isn't a bottleneck during build.	Should	A content spreadsheet/CMS-ready doc exists mapping 1 row per page: URL, title, meta description, body.

Phase 1 — Marketing Website

Epic 1.1 — Global Site Shell & Navigation

#	User Story	Pri.	Acceptance Criteria
1.1.1	As a visitor, I want a sticky header with logo, main nav (Products/Services/Locations/Gallery/Blog/Contact) and a visible phone/WhatsApp button, so I can act immediately from anywhere on the site.	Must	Header renders on all pages; nav collapses to a mobile hamburger under 768px; phone/WhatsApp links use tel:/wa.me deep links.
1.1.2	As a visitor, I want a footer with service areas, quick links, and social/contact info, so I can find secondary information without scrolling up.	Must	Footer present on all pages; locality list links to the relevant SEO pages.
1.1.3	As a visitor, I want the site to load fast and look correct on mobile, since most searches happen on phones.	Must	Lighthouse mobile performance score ≥ 85; responsive at 360/768/1024/1440px breakpoints.

Epic 1.2 — Homepage

#	User Story	Pri.	Acceptance Criteria
1.2.1	As a visitor, I want a hero section with a clear headline, sub-headline and CTA buttons (Get Free Quote / Book Site Visit / Emergency Service), so I immediately understand what Apollo Elevators offers.	Must	Hero renders above the fold; all 3 CTA buttons functional (form / booking / call link).
1.2.2	As a visitor, I want to see all 6 product categories with icons/images on the homepage, so I can jump to the one relevant to me.	Must	6 product cards displayed, each linking to its product detail page.

#	User Story	Pri.	Acceptance Criteria
1.2.3	As a visitor, I want an "Why Choose Apollo" trust-signals section (experience, safety standards, genuine parts, fast support), so I feel confident enquiring.	Should	Section renders with the 12 trust points from the brochure content, grouped into 3–4 columns.
1.2.4	As a visitor, I want to see a snippet of recent projects and testimonials on the homepage, so I have social proof before scrolling further.	Should	At least 3 project thumbnails and 2 testimonials shown, pulling from Gallery/Testimonials content.

Epic 1.3 — Product Pages (7 Lift Types)

#	User Story	Pri.	Acceptance Criteria
1.3.1	As a visitor, I want a dedicated page per lift type (Passenger, Home, Hospital, Freight, MRL, Capsule, Glass) with features, use-cases and images, so I can evaluate the right product for my building.	Must	7 pages live, each with unique title/meta/H1, feature list from brochure content, and a quote CTA.
1.3.2	As a visitor, I want an index page listing all products with quick-compare cards, so I can browse before drilling into one.	Should	/products page lists all 7 with a 1-line summary and link to each detail page.
1.3.3	As a visitor, I want each product page to have structured data (schema.org Product) so it can appear richly in Google search results.	Could	Valid JSON-LD present and passes Google's Rich Results Test for at least the Passenger Lift page.

Epic 1.4 — Service Pages (5 Services)

#	User Story	Pri.	Acceptance Criteria
1.4.1	As a visitor, I want a dedicated page each for Installation, AMC, Repair & Breakdown, Modernization and Inspection, describing what's included, so I understand the scope before enquiring.	Must	5 pages live with unique content per the brochure's service breakdown (e.g. AMC page lists all 9 AMC inclusions).
1.4.2	As a visitor, I want the AMC page to clearly explain what's covered vs. not, so I can compare it to other vendors.	Should	AMC inclusions rendered as a checklist; a short FAQ addresses 3 common questions.

Epic 1.5 — Location SEO Pages (~40 Pages)

#	User Story	Pri.	Acceptance Criteria
1.5.1	As a visitor searching for a locality-specific service, I want a page like "Lift Installation in Whitefield" so the result matches my exact search intent.	Must	A reusable page template exists; content is generated for all Bengaluru localities listed in the brochure (~18) × relevant services, or at minimum one combined locality page per area.
1.5.2	As a visitor outside Bengaluru, I want to see that Apollo Elevators serves my city (e.g. Mysuru, Mangaluru, Hubballi), so I know they can take on my project.	Should	A /locations page lists all Karnataka cities served, each with a short page or anchor section.
1.5.3	As the SEO owner, I want every locality page to have a unique meta title/description templated from {service} + {locality}, so pages don't get flagged as duplicate content.	Must	No two locality pages share an identical meta title/description; template uses dynamic variables.

Epic 1.6 — Projects & Gallery

#	User Story	Pri.	Acceptance Criteria
1.6.1	As a visitor, I want to browse completed projects filtered by building type (apartment, villa, hospital, commercial, industrial), so I can see relevant past work.	Should	Gallery page with filter tabs; each project shows images and a short description.
1.6.2	As a visitor, I want to view a lightbox/full-size image gallery, so I can inspect installation quality closely.	Could	Clicking a thumbnail opens a lightbox with next/prev navigation.

Epic 1.7 — Blog

#	User Story	Pri.	Acceptance Criteria
1.7.1	As a visitor, I want to read maintenance tips and safety articles, so I trust Apollo Elevators' expertise before enquiring.	Should	Blog index + at least 3 seed articles live, each with title/meta/author/date.

Epic 1.8 — Contact & Enquiry Capture

#	User Story	Pri.	Acceptance Criteria
1.8.1	As a visitor, I want a contact form (name, phone, email, city, building type, floors, message) on every relevant page, so I can request a quote without leaving the page.	Must	Form validates required fields client-side; submits via API; shows a success confirmation.
1.8.2	As the business owner, I want every form submission saved to the database and to trigger an email alert to sales, so no enquiry is missed.	Must	Submission persisted with timestamp, source page and UTM params; email sent within 1 minute of submission.
1.8.3	As a visitor, I want one-tap Call and WhatsApp buttons visible on mobile, so I can reach Apollo Elevators instantly for urgent breakdowns.	Must	Floating action buttons present on mobile viewport across all pages; wa.me link opens with a prefilled message.
1.8.4	As a visitor, I want to see the Bengaluru office location on an embedded map, so I know where Apollo Elevators is based.	Could	Google Map (or Leaflet, cost-free alternative) embed shows correct pin and address on the Contact page.

Epic 1.9 — SEO Infrastructure & Analytics

#	User Story	Pri.	Acceptance Criteria
1.9.1	As the SEO owner, I want an auto-generated XML sitemap and robots.txt, so search engines can crawl the whole site efficiently.	Must	sitemap.xml lists all live pages and updates automatically when new pages are added; robots.txt allows crawling of public pages.
1.9.2	As the business owner, I want Google Analytics/Search Console and a basic conversion pixel wired in, so I can see traffic and enquiry conversion.	Must	GA4 property receiving pageview events; Search Console verified; form submission fires a conversion event.
1.9.3	As the SEO owner, I want every page to support per-page meta title, description and OG image, so social shares and search snippets look correct.	Must	CMS/data model supports per-page SEO fields (built in Phase 1, editable via CMS from Phase 2).

9. Product Backlog — Phase 2 (Admin Panel & CMS)

Goal: Let Apollo Elevators staff run the website without a developer **Duration:** 3 weeks **Primary roles:** Sales/Admin, Manager

Epic 2.1 — Authentication & Roles

#	User Story	Pri.	Acceptance Criteria
2.1.1	As an admin, I want to log in with email/password (JWT-based), so I can access the admin panel securely.	Must	Login issues a JWT + refresh token; invalid credentials show a clear error; session persists across refresh.
2.1.2	As an admin, I want role-based access (Admin vs Manager) so sensitive actions (user management, deleting data) are restricted appropriately.	Must	Manager role cannot access User Management; attempting a restricted action returns 403 and is hidden in the UI.
2.1.3	As an admin, I want to reset my password via email if I forget it, so I'm not locked out.	Should	Forgot-password flow sends a time-limited reset link; link expires after 30 minutes.

Epic 2.2 — Dashboard

#	User Story	Pri.	Acceptance Criteria
2.2.1	As a manager, I want to see today's and this month's enquiry counts on login, so I have an immediate pulse on lead flow.	Must	Dashboard cards show today/7-day/30-day enquiry counts, refreshed on load.
2.2.2	As a manager, I want a simple chart of monthly leads and projects won, so I can track trends without exporting data.	Should	A line/bar chart renders the last 6 months of enquiry and won-project counts.

Epic 2.3 — Enquiry Management

#	User Story	Pri.	Acceptance Criteria
2.3.1	As sales staff, I want to see all website enquiries in a sortable, filterable table (by date, city, status), so I can prioritize follow-ups.	Must	Table supports filter by status/city and sort by date; pagination for >50 rows.
2.3.2	As sales staff, I want to open an enquiry, add notes, and change its status (New → Contacted → Quoted → Won/Lost), so the team has a shared view of pipeline state.	Must	Status change is persisted with a timestamp and the staff member who made it; notes are timestamped and attributable.
2.3.3	As sales staff, I want to assign an enquiry to a specific staff member, so accountability is clear.	Should	Assignee field selectable from active admin users; assigned enquiries appear in that user's "My Enquiries" view.
2.3.4	As sales staff, I want to mark an enquiry "Won" and have it create a Customer + Building + Lift record automatically, so I don't re-enter the same data in Phase 3's system.	Must	Marking Won opens a pre-filled Customer/Building creation form using the enquiry's data; confirming creates linked records.

Epic 2.4 — Content Management (Blog / Gallery / Testimonials)

#	User Story	Pri.	Acceptance Criteria
2.4.1	As an admin, I want to create, edit, publish/unpublish and delete blog posts with a rich text editor, so I can manage content without redeploying code.	Must	CRUD works end-to-end; draft posts are not visible on the public site; publish makes it live immediately.

#	User Story	Pri.	Acceptance Criteria
2.4.2	As an admin, I want to upload and categorize gallery images/videos by project type, so the public Gallery page stays current.	Must	Upload supports multi-file selection; images stored in S3; categorization tags selectable from a fixed list.
2.4.3	As an admin, I want to add/edit/remove customer testimonials with a star rating, so the homepage social proof stays fresh.	Should	Testimonials CRUD works; a "featured" flag controls homepage display.

Epic 2.5 — SEO Management

#	User Story	Pri.	Acceptance Criteria
2.5.1	As an admin, I want to edit meta title, description and OG image per page from the admin panel, so I can adjust SEO without a code change.	Must	Editing a page's SEO fields updates the live page's tags immediately.
2.5.2	As an admin, I want a simple checklist/status indicator showing which pages are missing a meta description, so I can close SEO gaps.	Could	Admin panel lists pages with missing/short (<50 char) meta descriptions.

Epic 2.6 — Project / Case Study Management

#	User Story	Pri.	Acceptance Criteria
2.6.1	As an admin, I want to add a completed project (type, location, images, description) so it appears in the public Projects & Gallery page.	Should	New project entries appear on the public gallery within the correct category filter.

Epic 2.7 — User Management

#	User Story	Pri.	Acceptance Criteria
2.7.1	As an admin, I want to create/deactivate admin panel users and assign roles, so staff access can be controlled as the team changes.	Should	Only Admin role can access this screen; deactivated users can no longer log in.

10. Product Backlog — Phase 3 (Service Management System)

Goal: Run day-to-day operations — customers, AMC, complaints, billing — from the software

Duration: 4 weeks

Primary roles: Manager, Sales/Admin

Epic 3.1 — Customer & Building Management

#	User Story	Pri.	Acceptance Criteria
3.1.1	As an admin, I want to create/edit/search customers with contact details and type (individual/builder/facility), so all customer data lives in one place.	Must	CRUD works; search by name/phone returns results in <1s for up to 5,000 records.
3.1.2	As an admin, I want to add one or more buildings under a customer (address, type, floors), so multi-property customers (builders) are modeled correctly.	Must	A customer can have N buildings; each building shows its lift count.

Epic 3.2 — Lift Management

#	User Story	Pri.	Acceptance Criteria
3.2.1	As an admin, I want to register a lift under a building with capacity, stops, manufacturer, controller type, install date and warranty end date, so every asset is trackable.	Must	Lift form validates required fields; lift auto-links to its building/customer.
3.2.2	As a manager, I want to see a lift's full history (AMC, complaints, visits, invoices) on one screen, so I don't have to cross-reference multiple modules.	Must	Lift detail page aggregates AMC status, last 10 visits, open complaints and invoice list.

Epic 3.3 — AMC Management

#	User Story	Pri.	Acceptance Criteria
3.3.1	As an admin, I want to create an AMC contract for a lift with start/end date and visits-per-year, so preventive maintenance is scheduled automatically.	Must	Creating an AMC auto-generates the visit schedule for its full term (e.g. 4 visits/year → 4 dated ServiceVisit records).
3.3.2	As the system, I want to send an automatic reminder to the customer and assigned engineer before each scheduled AMC visit, so visits aren't missed.	Must	Reminder (email/SMS) fires 2 days before each scheduled visit date via the Notification module.
3.3.3	As the system, I want to flag AMC contracts expiring within 30 days and notify the customer for renewal, so renewal revenue isn't lost.	Must	A scheduled job runs daily; contracts crossing the 30-day threshold trigger one renewal reminder (not repeated daily).
3.3.4	As a manager, I want a report of all AMC contracts by status (Active/Expiring/Expired), so I can follow up on renewals proactively.	Should	Report is filterable by status and date range; exportable to CSV.

Epic 3.4 — Complaint Management

#	User Story	Pri.	Acceptance Criteria
3.4.1	As an admin, I want to log a complaint against a specific lift with description, priority and optional photo, so breakdowns are tracked from first report.	Must	Complaint created with status=Pending; linked to the correct Lift/Customer.
3.4.2	As a manager, I want to assign a complaint to an available engineer, so field response is coordinated.	Must	Assignment creates a linked ServiceVisit and notifies the engineer; complaint status updates to Assigned.
3.4.3	As a manager, I want to see all open complaints sorted by age/priority, so nothing breaches SLA unnoticed.	Must	List view highlights complaints open >24 hours (configurable) in a warning color.

Epic 3.5 — Service Visit Scheduling

#	User Story	Pri.	Acceptance Criteria
3.5.1	As a manager, I want a calendar/list view of all scheduled visits per engineer per day, so I can balance workload.	Should	Visits grouped by engineer and date; drag-to-reassign or a reassign action is available.
3.5.2	As a manager, I want to reschedule a visit and have the customer/engineer notified of the new date, so changes don't cause confusion.	Should	Rescheduling updates the record and triggers a notification to both parties.

Epic 3.6 — Invoicing

#	User Story	Pri.	Acceptance Criteria
3.6.1	As an admin, I want to generate an invoice from an AMC contract or a completed service visit, so billing doesn't require a separate manual process.	Must	Invoice auto-populates customer, line items and amount; downloadable as PDF.
3.6.2	As an admin, I want to mark invoices Paid/Unpaid and see overdue invoices, so collections are tracked.	Must	Overdue filter shows invoices past due_date with status still Unpaid.

Epic 3.7 — Reports & Dashboards

#	User Story	Pri.	Acceptance Criteria
3.7.1	As a manager, I want a dashboard of AMC expiry, open complaints, and monthly revenue, so I can manage the business at a glance.	Should	Dashboard combines the three metrics on one screen, each linking through to the detailed list.
3.7.2	As a manager, I want an engineer performance report (visits completed, average resolution time), so I can review field team output.	Could	Report filterable by engineer and date range.

11. Product Backlog — Phase 4 (Engineer Portal)

Goal: Make field service paperless with a mobile-first engineer app **Duration:** 3 weeks **Primary roles:** Field Engineer

Epic 4.1 — Engineer Auth & Job List

#	User Story	Pri.	Acceptance Criteria
4.1.1	As an engineer, I want to log in on my phone with a simple username/password or OTP, so I can access my jobs without IT support.	Must	Login works on a 360px-wide mobile screen; session persists for the work day.
4.1.2	As an engineer, I want to see today's assigned jobs (AMC visits + complaints) as a simple list with customer name, address and job type, so I know my day's route.	Must	Job list loads in <2s on 4G; each job shows type, address and a status badge.
4.1.3	As an engineer, I want to tap a job to see full details and a one-tap "navigate" link (opens Google Maps), so I can get there without re-typing the address.	Must	Navigate link opens the maps app pre-filled with the building address.

Epic 4.2 — Visit Execution

#	User Story	Pri.	Acceptance Criteria
4.2.1	As an engineer, I want to mark "Visit Started" when I arrive, capturing a GPS timestamp, so there's an objective record of arrival time.	Must	GPS coordinates + timestamp saved on tap; works even if accuracy is approximate (falls back gracefully if permission denied).
4.2.2	As an engineer, I want a digital checklist (door, brake, motor, oil, controller, VVVF, ARD, cabin, alarm, lighting, emergency phone) that I can tick through per visit, so nothing is missed and there's a record of what was checked.	Must	Checklist items match the AMC inclusions from Section 5; each item supports OK/Needs Attention/NA plus an optional note.
4.2.3	As an engineer, I want to mark "Visit Completed" with a closing GPS timestamp, so total time on site is recorded automatically.	Must	Completion is blocked until the checklist has at least been reviewed (all items touched or explicitly skipped).

Epic 4.3 — Photo, Video & Signature Capture

#	User Story	Pri.	Acceptance Criteria
4.3.1	As an engineer, I want to take before/after photos directly in the app and attach them to the visit, so there's visual proof of work done.	Must	Camera capture or gallery upload works; photos upload to S3 and link to the visit record.
4.3.2	As an engineer, I want the customer to sign on my screen at job completion, and to add my own signature, so there's mutual sign-off with no paper.	Must	Signature pad captures a touch-drawn signature for both parties and stores it as an image attached to the visit.

Epic 4.4 — Service Report Generation

#	User Story	Pri.	Acceptance Criteria
4.4.1	As the system, I want to auto-generate a PDF service report (checklist results + photos + signatures) as soon as a visit is marked Completed, so no manual report writing is needed.	Must	PDF generated within 30 seconds of completion; matches a consistent Apollo Elevators branded template.

#	User Story	Pri.	Acceptance Criteria
4.4.2	As an engineer, I want the completed report automatically shared with the customer (email/WhatsApp) and synced back to the Admin Panel, so the office and customer both have it instantly.	**Must**	Report delivery is triggered automatically; delivery status is visible to admin (sent/failed).

Epic 4.5 — Offline Support (Stretch)

#	User Story	Pri.	Acceptance Criteria
4.5.1	As an engineer, I want to fill a checklist even with no signal (basement machine rooms often have none), with the data syncing once I'm back online, so poor connectivity doesn't block my work.	**Could**	Checklist entries are cached locally and sync automatically when connectivity resumes, with no data loss.

12. Product Backlog — Phase 5 (Customer Portal)

Goal: Reduce inbound calls by giving customers self-service visibility	**Duration:** 4 weeks	**Primary roles:** Building Customer / Facility Owner

Epic 5.1 — Customer Auth & Dashboard

#	User Story	Pri.	Acceptance Criteria
5.1.1	As a customer, I want to log in with my phone number/OTP or email, so I don't need to remember another password.	Must	OTP login works via SMS; session persists across visits on the same device.
5.1.2	As a customer, I want a dashboard showing my buildings/lifts, current AMC status and any open complaints, so I get the full picture on login.	Must	Dashboard aggregates data scoped only to the logged-in customer's own records.

Epic 5.2 — Asset & AMC Visibility

#	User Story	Pri.	Acceptance Criteria
5.2.1	As a customer, I want to see each of my lifts' installed date, warranty status and AMC validity, so I always know coverage without calling the office.	Must	Each lift card shows AMC status as Active/Expiring Soon/Expired with the exact end date.
5.2.2	As a customer, I want to see my full service visit history per lift with dates and engineer remarks, so I have a maintenance record.	Should	History list is paginated and sorted by most recent visit first.

Epic 5.3 — Complaint Raise & Track

#	User Story	Pri.	Acceptance Criteria
5.3.1	As a customer, I want to raise a complaint by selecting the affected lift, describing the issue and optionally attaching a photo, so I don't have to call for routine breakdowns.	Must	Submitted complaint appears immediately in the Admin Panel's complaint queue with status Pending.
5.3.2	As a customer, I want to track my complaint's status (Pending → Assigned → In Progress → Completed) in real time, so I know when to expect the engineer.	Must	Status updates made by admin/engineer reflect in the customer's view within a few seconds of refresh/poll.

Epic 5.4 — AMC Renewal & Online Payment

#	User Story	Pri.	Acceptance Criteria
5.4.1	As a customer, I want to renew an expiring AMC directly from the portal and pay online, so renewal doesn't require a phone call or bank transfer.	Should	Renewal flow creates a new AMC cycle on successful payment via the integrated payment gateway (e.g. Razorpay).
5.4.2	As a customer, I want to download a payment receipt immediately after paying, so I have proof for my records.	Should	Receipt PDF is generated and downloadable/emailed right after a successful transaction.

Epic 5.5 — Documents & Downloads

#	User Story	Pri.	Acceptance Criteria
5.5.1	As a customer, I want to download past invoices, AMC certificates and service reports as PDFs, so I don't need to email the office for paperwork.	Must	All documents linked to the customer's account are listed and downloadable in one place.

Epic 5.6 — Notifications & Feedback

#	User Story	Pri.	Acceptance Criteria
5.6.1	As a customer, I want to receive SMS/WhatsApp/email updates when my complaint status changes or an AMC visit is scheduled, so I stay informed without checking the portal.	Should	Notification fires on each relevant status transition, using the customer's preferred channel if set.
5.6.2	As a customer, I want to rate a completed service visit (1–5 stars + comment), so Apollo Elevators can track service quality.	Could	Feedback prompt appears after a visit is marked Completed; rating is stored against the engineer and visit.

13. Definition of Done, NFRs & Working Agreement

Definition of Done (applies to every user story)

- Code merged to `develop` via pull request, reviewed by at least one other person (or self-reviewed with a checklist if solo).
- Acceptance criteria in the backlog table are verified manually or via an automated test.
- No console errors; API returns proper HTTP status codes and validation error messages.
- Responsive on mobile for any customer/engineer-facing screen (this covers the majority of Phase 1, 4 and 5).
- Environment variables / secrets are not hard-coded; config is externalized.
- Feature deployed to staging and smoke-tested before being marked Done.

Non-Functional Requirements

Category	Requirement
Performance	Public pages: Lighthouse mobile score ≥ 85. API p95 response time < 500ms for standard CRUD endpoints.
Security	JWT with short-lived access tokens + refresh tokens; passwords hashed with bcrypt/argon2; role-based authorization enforced server-side (never trust the frontend alone).
Availability	Target 99.5% uptime once Phase 3+ is live and business-critical; automated daily PostgreSQL backups with a tested restore process.
Data Privacy	Customer PII (phone, email, address) restricted to authorized roles; portal users can only ever see their own records.
Auditability	Every status change (enquiry, complaint, AMC, invoice) is timestamped and attributed to a user, stored for at least 2 years.
Accessibility	Public site meets basic WCAG 2.1 AA essentials: color contrast, alt text on images, keyboard-navigable forms.
Browser/Device Support	Latest 2 versions of Chrome, Safari, Edge; iOS Safari and Android Chrome for the Engineer PWA and Customer Portal.

Suggested Working Rhythm

- Treat each Epic in Sections 8–12 as a sprint-sized chunk of work (roughly 3–10 days for one developer).
- Work top-to-bottom within a phase, but **Must**-priority stories in every epic take precedence over **Should**/**Could** ones.
- Re-prioritize freely — this backlog is a complete reference, not a rigid order. Pull whichever story unblocks the business fastest.
- Each Phase in Section 7 is a real release: deploy and start using it before moving to the next phase's development.

End of document. This blueprint is derived from the Apollo Elevators product brochure and prior technical planning discussion, restructured into a sitemap, architecture, workflows and a sprint-ready backlog for independent, phase-by-phase development.